

First Assignment

5.9.2024

5/5 Points

Forsøk 1 



Se gjennom tilbakemelding

5.9.2024

Forsøk 1 Poengsum:

5/5



Legg til kommentar

Anonym vurdering: **nei**

3 tillatte forsøk

29.8.2024 til 6.9.2024

▼ Detaljer

Playlist Management System

Objective:

This assignment aims to apply the concept of a Doubly Linked List to create a Playlist Management System.

Implement a Doubly Linked List to manage a playlist of songs, allowing for efficient **insertion**, **deletion**, and **traversal** in both directions.

Playlist Management System Description:

Create a system to manage a playlist of songs. Each song will have a unique **ID**, **title**, and **artist**. The system should allow users to add songs, remove songs, and navigate through the playlist forward and backward.

Task 1: Implementation of Doubly Linked List (3 points)

Create a C++ class named Playlist to represent the doubly linked list for the playlist management system.

The class should have private member variables to hold the data (song ID, title, artist) and pointers to the next and previous nodes.

Implement the constructor(s) and destructor to manage memory allocation and deallocation properly.

Task 2: Playlist Management Simulation (2 points)

In the main() function, create an instance of the Playlist class.

Add a given number of songs (e.g., 5, 10, or 15) to the playlist by taking input from the user for each song's ID, title, and artist.

Provide a menu-driven interface for the user to perform the following operations:

- Add a new song by taking input from the user.
- Remove a song by taking the song ID as input from the user.
- Display the playlist.
- Display the playlist in reverse order.
- Search a song

Submission Guidelines:

- Organize the code neatly with proper indentation and meaningful variable names.
- Include comments in the code to explain complex logic and functions.
- Provide a readme file describing the Playlist Management System and the approach used.
- Zip the code files, readme, and any additional resources into a single archive and submit it on the assignment portal by the due date.

Note: You are not allowed to use multiple files or built-in libraries.
